Titanic Project

1 Loads The Data Set

```
import pandas as pd
```

```
df = pd.read_csv('titanic.csv')
```

```
df.head(12)
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S
5	6	0	3	Moran, Mr. James	male	NaN	0	0	330877	8.4583	NaN	Q
6	7	0	1	McCarthy, Mr. Timothy J	male	54.0	0	0	17463	51.8625	E46	S
7	8	0	3	Palsson, Master. Gosta Leonard	male	2.0	3	1	349909	21.0750	NaN	S
8	9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27.0	0	2	347742	11.1333	NaN	S
9	10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	female	14.0	1	0	237736	30.0708	NaN	C
10	11	1	3	Sandstrom, Miss. Marguerite Rut	female	4.0	1	1	PP 9549	16.7000	G6	S
11	12	1	1	Bonnell, Miss. Elizabeth	female	58.0	0	0	113783	26.5500	C103	S

Next steps: Generate code with df New interactive sheet

2 EDA Processing

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
```

```
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
df.isnull().sum()
```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

columnu name Age missing value replace with mean()

```
df['Age'] = df['Age'].fillna(df['Age'].mean())
```

```
df['Age'].isnull().sum()
```

```
np.int64(0)
```

columnu name Embarked missing value replace with mode()

```
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])
```

```
df['Embarked'].isnull().sum()
```

```
np.int64(0)
```

Drop The Column we don't want

```
df = df.drop(['PassengerId','Name','Ticket','Cabin'],axis=1)
```

```
df.head()
```

	Survived	Pclass	Sex	Age	SibSp	Parch	Fare	Embarked
0	0	3	male	22.0	1	0	7.2500	S
1	1	1	female	38.0	1	0	71.2833	C
2	1	3	female	26.0	0	0	7.9250	S
3	1	1	female	35.0	1	0	53.1000	S
4	0	3	male	35.0	0	0	8.0500	S

Next steps: Generate code with df New interactive sheet

column name Sex

Male = 1

Female = 0

```
df['Sex'] = df['Sex'].map({'male':1,'female':0})
```

```
df.head()
```

	Survived	Pclass	Sex	Age	SibSp	Parch	Fare	Embarked
0	0	3	1	22.0	1	0	7.2500	S
1	1	1	0	38.0	1	0	71.2833	C
2	1	3	0	26.0	0	0	7.9250	S
3	1	1	0	35.0	1	0	53.1000	S
4	0	3	1	35.0	0	0	8.0500	S

Next steps: Generate code with df New interactive sheet

```
df['Embarked'].unique()
```

```
array(['S', 'C', 'Q'], dtype=object)
```

```
df['Embarked'] = df['Embarked'].map({'S':1,'C':2,'Q':3})
```

```
df.head()
```

	Survived	Pclass	Sex	Age	SibSp	Parch	Fare	Embarked
0	0	3	1	22.0	1	0	7.2500	1
1	1	1	0	38.0	1	0	71.2833	2
2	1	3	0	26.0	0	0	7.9250	1
3	1	1	0	35.0	1	0	53.1000	1
4	0	3	1	35.0	0	0	8.0500	1

Next steps: Generate code with df New interactive sheet

3 Divided Data sets into Independent & Dependent Variables

```
x = df.drop(['Survived'],axis=1)
y = df['Survived']
```

```
x.head()
```

	Pclass	Sex	Age	SibSp	Parch	Fare	Embarked
0	3	1	22.0	1	0	7.2500	1
1	1	0	38.0	1	0	71.2833	2
2	3	0	26.0	0	0	7.9250	1
3	1	0	35.0	1	0	53.1000	1
4	3	1	35.0	0	0	8.0500	1

Next steps: Generate code with x New interactive sheet

```
y.head()
```

	Survived
0	0
1	1
2	1
3	1
4	0

dtype: int64

4 Train Test Split

```
from sklearn.model_selection import train_test_split
```

```
x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=0.2,random_state=42)
```

5 Training The Model with Logistic Regression

```
import warnings
warnings.filterwarnings('ignore')
```

```
from sklearn.linear_model import LogisticRegression
```

```
model = LogisticRegression()
```

```
model.fit(x_train,y_train)
```

```
LogisticRegression
LogisticRegression()
```

```
y_pred = model.predict(x_test)
y_pred
```

```
array([0, 0, 0, 1, 1, 1, 1, 0, 1, 1, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0, 0, 0,
       1, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 1, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0,
       1, 0, 0, 0, 0, 0, 1, 1, 0, 1, 0, 1, 0, 1, 1, 1, 0, 1, 1, 0, 0, 1,
       0, 0, 0, 1, 1, 1, 1, 1, 0, 0, 1, 1, 1, 0, 0, 1, 1, 0, 0, 0, 1, 1,
       0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0,
       1, 0, 1, 0, 1, 0, 0, 0, 1, 0, 0, 1, 1, 0, 0, 0, 1, 1, 1, 0, 1, 0,
       0, 1, 0, 1, 1, 0, 0, 1, 0, 1, 0, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 1,
       0, 0, 0, 1, 1, 1, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 1, 1, 0, 1, 0, 0,
       0, 1, 1])
```

```
# Actually data
y_test
```

	Survived
709	1
439	0
840	0
720	1
39	1
...	...
433	0
773	0
25	1
84	1
10	1

179 rows × 1 columns

dtype: int64

Accuracy of Mode

```
from sklearn.metrics import accuracy_score
```

```
accuracy_score(y_pred, y_test)
```

```
0.7988826815642458
```

```
model.score(x_test,y_test)
```

```
0.7988826815642458
```

```
model.predict([[3,1,22.0,1,0,7.2500,1]])
```

```
array([0])
```

6 Dump The File with help of Pickle

```
import pickle
```

```
pickle.dump(model,open('titanic_model.pkl','wb'))
```

```
Start coding or generate with AI.
```